

ECON 300 – Labour Economics

Step-by-Step Solutions: Quiz on Chapters 11 and 16

Immigration and Unemployment

Question 1: Immigration Composition and Selection Policy

Total immigrants:

420,000

Categories:

- Economic immigrants: 189,000
- Family-class immigrants: 147,000
- Refugees and humanitarian immigrants: 84,000

(a) Percentage share of each category

The formula is:

$$\text{Percentage share} = \frac{\text{category size}}{\text{total immigration}} \times 100$$

Economic immigrants:

$$\frac{189,000}{420,000} \times 100 = 45\%$$

45%

Family-class immigrants:

$$\frac{147,000}{420,000} \times 100 = 35\%$$

35%

Refugees and humanitarian immigrants:

$$\frac{84,000}{420,000} \times 100 = 20\%$$

20%

(b) New category sizes and new total

Economic immigrants increase by 18%:

$$189,000(1.18) = 223,020$$

$$\boxed{223,020}$$

Family-class immigrants decrease by 8%:

$$147,000(1 - 0.08) = 147,000(0.92) = 135,240$$

$$\boxed{135,240}$$

Refugees and humanitarian immigrants unchanged:

$$84,000$$

$$\boxed{84,000}$$

New total number of immigrants:

$$223,020 + 135,240 + 84,000 = 442,260$$

$$\boxed{442,260}$$

(c) New percentage share of economic immigrants

$$\frac{223,020}{442,260} \times 100 \approx 50.42\%$$

$$\boxed{50.42\%}$$

Question 2: Unemployment, Participation, and Employment Rates

Working-age population:

6,800,000

Labour force participation rate:

67%

Unemployment rate:

6.5%

(a) Size of the labour force

$$\text{Labour force} = 0.67 \times 6,800,000$$

$$= 4,556,000$$

4,556,000

(b) Number of unemployed workers

$$\text{Unemployed} = 0.065 \times 4,556,000$$

$$= 296,140$$

296,140

(c) Number of employed workers

$$\text{Employed} = \text{Labour force} - \text{Unemployed}$$

$$= 4,556,000 - 296,140 = 4,259,860$$

4,259,860

(d) Employment rate

$$\begin{aligned}\text{Employment rate} &= \frac{\text{Employed}}{\text{Working-age population}} \times 100 \\ &= \frac{4,259,860}{6,800,000} \times 100 \approx 62.65\%\end{aligned}$$

$$\boxed{62.65\%}$$

Now suppose during a recession:

- 52,000 employed workers lose jobs and begin searching
- 36,000 unemployed workers become discouraged and leave the labour force
- 18,000 people previously out of the labour force enter and immediately begin searching

(e) New number of unemployed workers

Start with initial unemployed:

$$296,140$$

Add workers who lose jobs and begin searching:

$$296,140 + 52,000 = 348,140$$

Subtract unemployed workers who become discouraged:

$$348,140 - 36,000 = 312,140$$

Add new entrants who begin searching:

$$312,140 + 18,000 = 330,140$$

$$\boxed{330,140}$$

(f) New labour force

Start with initial labour force:

$$4,556,000$$

Workers who become discouraged leave the labour force:

$$4,556,000 - 36,000 = 4,520,000$$

People entering and searching join the labour force:

$$4,520,000 + 18,000 = 4,538,000$$

Note: workers who lose jobs but continue searching remain in the labour force, so they do not change the total labour force.

$$\boxed{4,538,000}$$

(g) New unemployment rate

$$\text{New unemployment rate} = \frac{330,140}{4,538,000} \times 100$$

$$\approx 7.27\%$$

$$\boxed{7.27\%}$$

Interpretation:

The unemployment rate rises from 6.5% to 7.27%, showing a deterioration in labour market conditions. However, it still does not fully reflect hardship because some discouraged workers left the labour force and are no longer counted as unemployed.

Question 3: Long-Term Unemployment and Labour Market Interpretation

Each country has a labour force of:

12,500,000

Unemployment rate in both countries:

5.6%

(a) Number of unemployed workers in each country

$$\text{Unemployed} = 0.056 \times 12,500,000$$

$$= 700,000$$

So each country has:

700,000

(b) Number of long-term unemployed workers

Country A:

$$0.14 \times 700,000 = 98,000$$

98,000

Country B:

$$0.41 \times 700,000 = 287,000$$

287,000

(c) Long-term unemployment rate as a percentage of the labour force

Formula:

$$\text{Long-term unemployment rate} = \frac{\text{Long-term unemployed}}{\text{Labour force}} \times 100$$

Country A:

$$\frac{98,000}{12,500,000} \times 100 = 0.784\%$$

0.78%

Country B:

$$\frac{287,000}{12,500,000} \times 100 = 2.296\%$$

2.30%

(d) Why Country B may still have a more serious labour market problem

Although both countries have the same overall unemployment rate of 5.6%, Country B has a much larger share of long-term unemployed workers. This suggests more serious labour market difficulties because long-term unemployment often leads to:

- skill deterioration,
- weaker re-employment prospects,
- greater financial hardship,
- more persistent structural problems.

So Country B likely faces a more severe unemployment problem even though the headline unemployment rate is the same.

Country B has the more serious labour market problem because unemployment is more persistent.

Final Answers Summary

Question 1

- (a) 45%, 35%, 20%
- (b) 223,020, 135,240, 84,000, 442,260
- (c) 50.42%

Question 2

- (a) 4,556,000
- (b) 296,140
- (c) 4,259,860
- (d) 62.65%
- (e) 330,140
- (f) 4,538,000
- (g) 7.27%

Question 3

- (a) 700,000 in each country
- (b) 98,000 in A, 287,000 in B
- (c) 0.78% in A, 2.30% in B